

PRIME PALINDROME

Java Program Documentation

1. Question

Write a Java program which reads a number and check whether the given number is a prime palindrome or not.

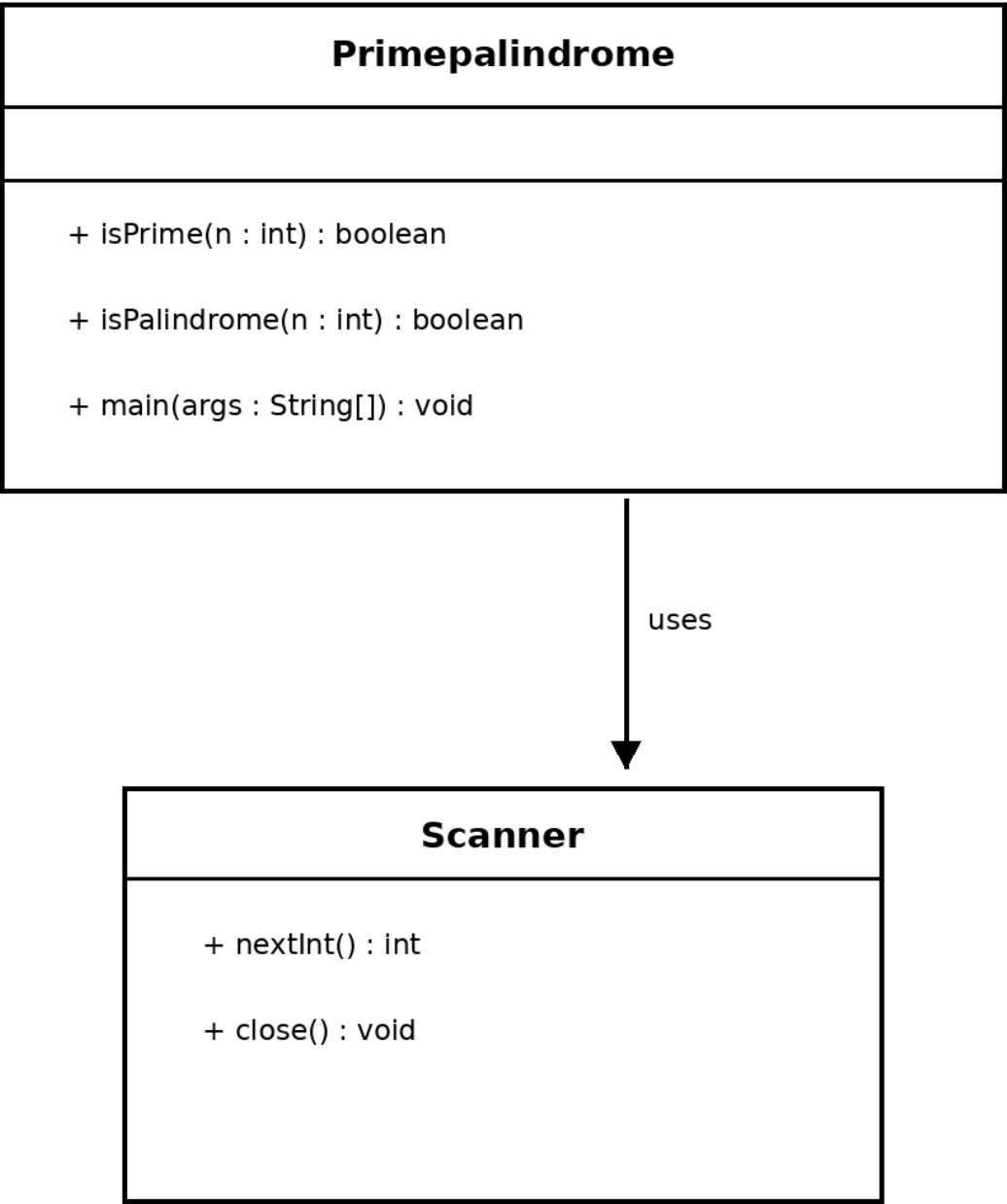
2. Execution Steps

1. Open Eclipse IDE or a terminal.
2. Create a Java project/package named Excercise.
3. Create a Java class named Primepalindrome.
4. Copy the complete program into Primepalindrome.java.
5. Make sure the public class name is Primepalindrome and the file name is Primepalindrome.java.
6. Save the program.
7. Compile the program.
8. Run the Primepalindrome class.
9. Enter a number when prompted.
10. Observe whether the number is a prime palindrome.

3. Algorithm

1. Start.
2. Define the Primepalindrome class.
3. Define the isPrime(int n) method to check whether n is prime.
4. If n is less than or equal to 1, return false.
5. Check divisibility from 2 to $n/2$. If n is divisible by any value, return false; otherwise return true.
6. Define the isPalindrome(int n) method.
7. Store the original number and reverse its digits.
8. Compare the original number with the reversed number. If they are equal, return true; otherwise return false.
9. Create a Scanner object and read a number from the user.
10. Check isPrime(n) and isPalindrome(n).
11. If both conditions are true, print that the number is a prime palindrome.
12. Otherwise, print that the number is not a prime palindrome.
13. Stop.

4. UML Class Diagram



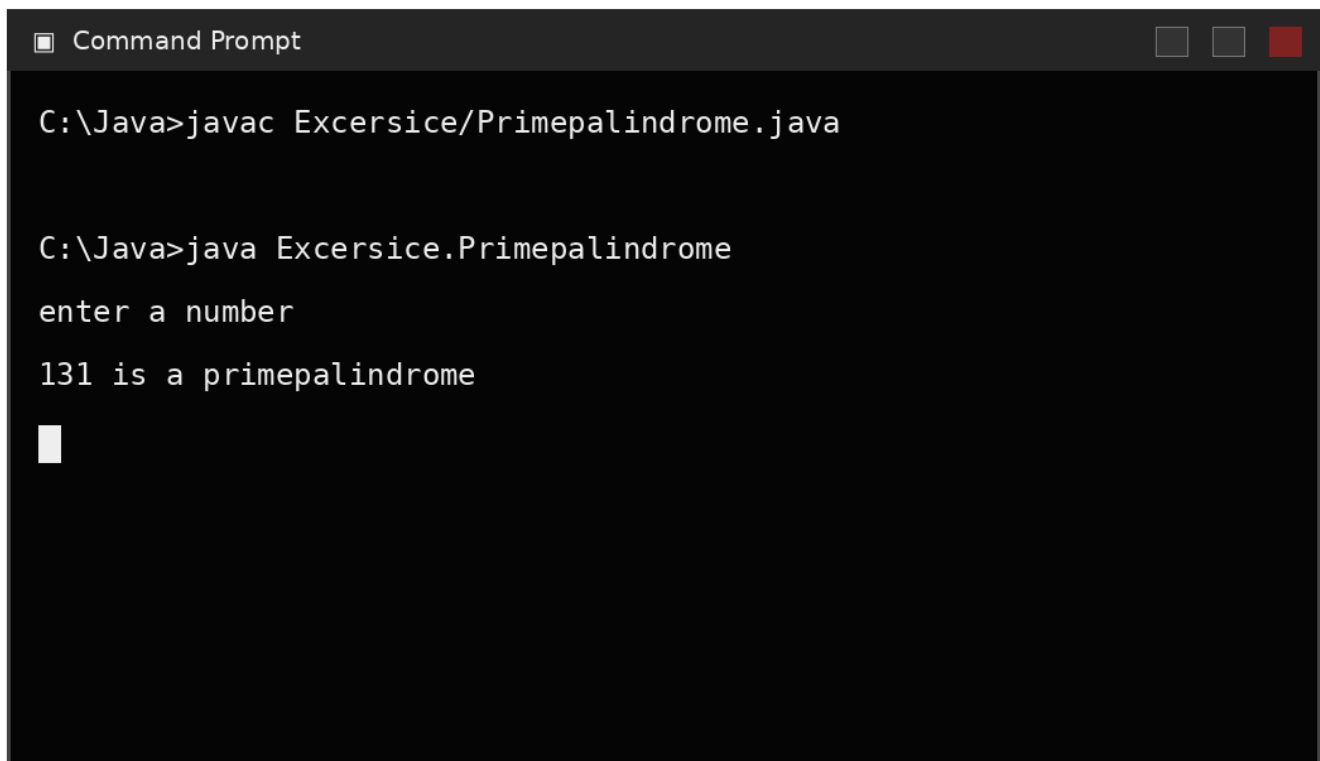
5. Java Source Code

```
package Excercise;
import java.util.Scanner;
public class Primepalindrome {
    static boolean isPrime(int n) {
        if(n<=1) {
            return false;
        }
        for(int i=2;i<=n/2;i++) {
            if(n%i==0) {
                return false;
            }
        }
        return true;
    }

    static boolean isPalindrome(int n) {
        int original = n;
        int reverse = 0;
        while(n != 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }
        if(original == reverse) {
            return true;
        }
        return false;
    }

    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        System.out.println("enter a number");
        int n = s.nextInt();
        if (isPrime(n) && isPalindrome(n)) {
            System.out.println(n + " " + "is a primepalindrome");
        }else {
            System.out.println(n + " " + "is not a primepalindrome");
        }
    }
}
```

6. Output – Terminal Screenshot



```
Command Prompt

C:\Java>javac Excercise/Primepalindrome.java

C:\Java>java Excercise.Primepalindrome
enter a number
131 is a primepalindrome
█
```

The screenshot shows a Windows Command Prompt window titled "Command Prompt". The user has navigated to the directory C:\Java. They first compile the Java file "Excercise/Primepalindrome.java" using the "javac" command. Then, they run the compiled class "Excercise.Primepalindrome" using the "java" command. The program prompts the user to "enter a number". The user has entered "131", and the program outputs "131 is a primepalindrome". A white cursor is visible on the line following the output.